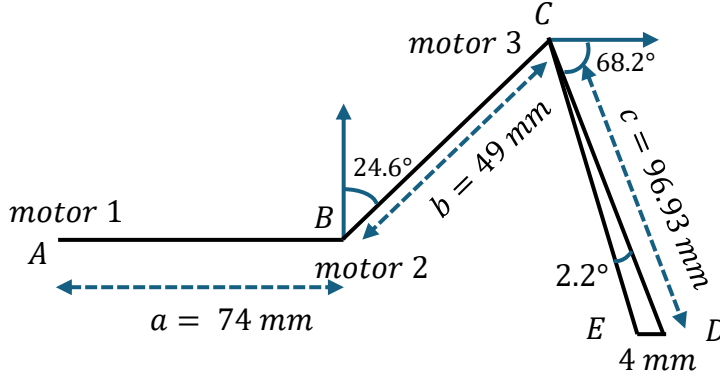


The inverse kinematics of the bot involves solving for how much the motors should move (l, m, n) in order for the tip to reach point $Y (x_2, y_2, z_2)$ from point $X (x_1, y_1, z_1)$ with l, m and n being the rotations of motor 1, 2 and 3 from their base positions respectively. The figure below demonstrates the constraints of the leg:



The positions of the motors (A, B and C) are determined by their centres. The blue arrows represent the direction of the motors w.r.t the side with which its angle has been given.

In order to determine (l, m, n) in terms of (x_1, y_1, z_1) and (x_2, y_2, z_2) we will take a coordinate system with the origin at A , x – axis along the initial direction of AB , y – axis perpendicular to AB and z axis out of the plane (plane being the xy – plane). We will ignore the length DE for the sake of calculations, i.e. $CD \approx CE$ although $CE = 95.52 \text{ mm}$ the value of c can be changed according to the orientation of the bot for better calculations.

Take the direction of rotation of motor 1 to be positive along the anticlockwise direction in the xz – plane and the directions of rotation of motors 2 and 3 to be positive along the clockwise direction in the xy – plane. We will measure the angle of motor 2 from the y – axis and the angle of motors 1 and 3 from the x – axis.

Since initial position of AB is along x – axis, we have $z_1 = 0$.

Let B', C', D' and E' be the points B, C, D and E after the tip has reached from point X to point Y . Consider the triangle formed by the points $A, (x_2, 0, 0)$ and $(x_2, 0, z_2)$. From here we have $\angle A = l = \text{atan2}(-z_2, x_2)$.

Now let angular position of BC from the y – axis is x and the angular position of CD from the x – axis is y . Hence the angular position of $B'C'$ from the y – axis is $x + m$ and the angular position of $C'D'$ from the x – axis is $y + m + n$.

Consider the triangle $B'C'E'$. Notice that $\angle B'C'E' = \frac{\pi}{2} + x - y - n$. So by cosine rule, if R is the length of $B'E'$ then,

$$n = \frac{\pi}{2} + x - y - \text{acos}\left(\frac{b^2 + c^2 - R^2}{2bc}\right)$$

Where $R = \sqrt{\left(\sqrt{x_2^2 + z_2^2} - a\right)^2 + y_2^2}$

And $\angle C'B'D' = g = \arccos\left(\frac{b^2 + R^2 - c^2}{2bR}\right)$

Consider the triangle $AB'D'$. By cosine rule, we have

$$\angle AB'D' = t = \arccos\left(\frac{a^2 + R^2 - x_2^2 - y_2^2 - z_2^2}{2aR}\right)$$

Notice that,

$$\angle AB'C' + \angle AB'D' + \angle C'B'D' = 2\pi$$

$$\frac{\pi}{2} + x + m + g + t = 2\pi$$

Hence, $m = \frac{3\pi}{2} - x - g - t$